Sebastian De Leon

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Objective

Bilingual **Junior in Computer Engineering (3.89 GPA)** with 2 years of experience in full-stack software engineering and embedded systems, seeking a **Summer 2027 Internship** in Software Engineering, Embedded Systems, or Robotics. Proven collaborator in multidisciplinary engineering teams, delivering production cloud applications, electromechanical control systems, and cross-platform software. **Open to relocation nationwide.**

Education

Western Michigan University — Expected Dec 2027
B.S. in Computer Engineering, GPA: 3.89 | Dean's List (All Semesters) — Kalamazoo, MI

- **Relevant Coursework**: Printed Circuit Assembly, MEMs fabrication, Embedded Systems, Data Structures & Algorithms, Digital Desing, Circuit Analysis

Experience

Junior Software Engineer — Jul 2024 – Jul 2026
Reispsolutions S.R.L. — Santo Domingo, Dom. Rep.

- Architected a distributed Learning Management System (LMS) in ASP.NET alongside senior engineers, standardizing employee onboarding and scaling training operations across dozens of multi-brand restaurant locations.
- Engineered production .NET 9 backend services using Command Query Responsibility Segregation (CQRS), Domain-Driven Design (DDD), and Entity Framework Core, conducting peer code reviews to ensure scalable architecture.
- Developed an end-to-end clinical management web platform that automated patient intake workflows, calendar scheduling, and SMS/email appointment reminders, significantly reducing manual front-desk coordination.
- Collaborated on cross-functional teams to deliver self-service Android tablet kiosk applications using Jetpack Compose, streamlining in-store customer ordering across multiple restaurant brands.

Manufacturing & Telemetry Assistant — Aug 2024 – Apr 2025
INTEC NASA Human Exploration Rover Challenge (HERC) Team — Santo Domingo, Dom. Rep.

- Partnered with mechanical and electrical sub-teams to design and fabricate the 2025 rover's liquid-detection payload: a stepper-motor lead-screw actuator that vertically positioned a sensor to verify water samples.
- Programmed the ESP32-S3 embedded rover control system, decoding Pulse-Width Modulation (PWM) signals from a radio-control (RC) receiver into a real-time state machine governing drivetrain motion and task execution.
- Directed full-stack development of the official Apolo 27 team web platform and coordinated a student developer team to build interactive digital games for national STEM outreach across the Dominican Republic.

Projects

Velo: Movie Matchmaker — Sep 2025 – Present
Shipped Cross-Platform Mobile Application — React Native, TypeScript, Supabase

- Develop a production React Native (Expo) mobile application that resolves group movie-night indecision through real-time multiplayer sessions, synchronized swipe decks, and ranked watchlists powered by weighted voting and veto algorithms.
- Maintain a scalable Clean Architecture codebase using TypeScript, Zustand, and Supabase (PostgreSQL, Realtime, Edge Functions) to securely proxy third-party metadata APIs, cache responses, and synchronize live session states.
- Integrate production features including QR/code session joining, guest-to-account data migration, Letterboxd watchlist imports, five-language localization, and RevenueCat/AdMob monetization.

Printune — May 2025 – Aug 2025
3D-Printable Spotify Keychain Generator — Next.js, TypeScript, Three.js, Python

- Created a full-stack web and desktop application converting Spotify track, album, and playlist links into custom, 3D-printable keychain models ready for slicing and additive manufacturing.
- Implemented client-side image processing to extract Spotify Code bar geometry and utilized Three.js (React Three Fiber) to render interactive 3D previews with customizable dimensions, colors, and extrusion profiles.
- Automated 3D mesh generation and stereolithography (STL) file export supporting custom 3D model uploads, enabling users to directly download print-ready files from the browser.

Quadruped Robot Dog — May 2026
Custom Embedded Robotics Prototype — ESP32-S3, C++, FreeCAD, PCB Design

- Designed and fabricated a 3D-printed quadruped robot chassis in FreeCAD, selecting electromechanical actuators and assembling custom embedded power and control circuitry on a perfboard.
- Engineered an embedded Wi-Fi web server on an ESP32-S3 microcontroller and programmed sequential servo gait kinematics to enable stable real-time smartphone control of walking and turning maneuvers.

Technical Skills

- **Programming Languages & Tools**: C/C++, C#, Python, TypeScript, JavaScript, SQL, Git, Docker, Linux, Supabase, PostgreSQL
- **Software & Frameworks**: .NET Core (ASP.NET, Entity Framework), React, React Native, Next.js, Jetpack Compose, Unity, Three.js
- **Embedded & Hardware**: ESP-IDF (ESP32), Arduino, OrCAD X, KiCad, LTspice, SolidWorks, Fusion 360, FreeCAD, 3D Printing